Three of the four sp^3 hybridised orbitals contains single electron and the fourth orbital is empty. Two of the half filled hybridised orbitals of each boron overlap with the 1s orbitals of two hydrogens to form four terminal 2c-2e bonds, leaving one empty and one half filled hybridised orbitals on each boron. The Three centre - two electron bonds), B-H-B bond formation involves overlapping the half filled hybridised orbital of one boron, the empty hybridised orbital of the other boron and the half filled 1s orbital of hydrogen.

Uses of diborane:

1. Diborane is used as a high energy fuel for propellant
2. It is used as a reducing agent in organic chemistry
3. It is used in welding torches

2.2.7 Boron trifluoride:

Preparation:

Boron trifuloride is obtained by the treatment of calcium fluoride with boron trioxide in presence of conc. sulphuric acid.

$$B_2O_3 + 3CaF_2 + 3H_2SO_4 \xrightarrow{\Delta} 2BF_3 + 3CaSO_4.+ 3H_2O$$

It can also be obtained by treating boron trioxide with carbon and fluorine.

$$B_2O_3 + 3C + 3F_2 \longrightarrow 2BF_3 + 3CO$$

In the laboratory pure BF_3 is prepared by the thermal decomposition of benzene diazonium tetrafluoro borate.

$$C_6H_5N_2BF_4 \xrightarrow{\Delta} BF_3 + C_6H_5F + N_2$$

Properties:

Boron trifluoride has a planar geometry. It is an electron deficient compound and accepts electron pairs to form coordinate covalent bonds. They form complex of the type $[BX_4]^-$.

$$BF_3 + NH_3 \longrightarrow F_3B \leftarrow NH_3$$
$$BF_3 + H_2O \longrightarrow F_3B \leftarrow OH_2$$

On hydrolysis, boric acid is obtained. This then gets converted into Hydro fluoroboric acid.

$$4BF_3 + 12H_2O \longrightarrow 4H_3BO_3 + 12HF$$
$$3H_3BO_3 + 12HF \longrightarrow 3H^+ + 3[BF_4]^- + 9H_2O$$
$$\overline{4BF_3 + 3H_2O \longrightarrow H_3BO_3 + 3H^+ + 3[BF_4]^-}$$

Uses of Boron trifluoride:

1. Boron trifluoride is used for preparing HBF_4, a catalyst in organic chemistry
2. It is also used as a fluorinating reagent.

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